UNIVERSITY OF CAMBRIDGE

MINI PROJECT1 LITERATURE REVIEW

UNIVERSITY OF CAMBRIDGE

DEPARTMENT OF CHEMICAL ENGINEERING AND BIOTECHNOLOGY

SENSOR CDT

Biomimetic flows to accelerate de-risking of nanocarrier drug delivery technologies

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1 Abstract

The clinical translation of nanocarrier-based targeted drug delivery is hindered by the inability of conventional static in-vitro models to simulate nanoparticle dynamic behavior, such as shear-induced aggregation and structural instability. This literature review analyses the causes of these bottlenecks and compares existing in-vitro blood circulation models, identifying critical gaps in nanoparticle stability assessment. Drawing on prior applications of microfluidics in drug development, microfluidic circulation platforms are identified as a promising strategy to reproduce physiologically relevant flow conditions and improve nanoparticle validation.

2 Main body

2.1 Introduction

In this literature review, nanocarrier in targeted drug delivery and its current bottleneck of clinical translation are explored explicitly. In-vitro microfluidic circulation method, which mimics in-vivo blood circulation, is discussed as a potential solution . Section one introduces the fundamental principles of nanocarrier-based drug delivery, and emphasizes the challenges of translating it into clinical use. It is shown that conventional static in vitro models are unable to predict in vivo performance. In section two, existing equipment that mimics the blood circulation to study the nanoparticle stability are reviewed. In section three, microfludics approach are identified and compared with other circulation-mimicking systems.

2.2 Nanocarrier Drug Delivery and Challenges

Nanotechnology is the intentional engineering and manipulation of particulate matter into a physical state that can be reassembled into nano-systems with improved functions [1]. Due to the advancement of nanotechnology, nanoparticles (NPs) as shown in Fig. 1. has been designed as carriers to accurately deliver drugs to specific tissues in the body [2]. NPs range in the size from 70 nm to 500 nm and have specific surface properties, which enables them to select diseased cells precisely while circulating in the blood vessels. When the cancer cells have been identified, nps diffuse and releas the therapeutic drug responsively to kill the selected cells [3]. Therefore, this target drug delivery method prevents healthy cell from damage and improves the drug efficacy for complex diseases.

However, current clinical translation of nanocarrier-based drug delivery systems is held back by the obstacles during blood circulation before NPs reach the target. NPs are transported within a strongly dynamic fluid environment, which is influenced by the forces generated by blood flow, commonly described as shear stress (SS). SS is the force per unit area created when a tangential force acts on the vessel wall and varies significantly with vessel geometry and flow velocity [4]. Under such flow conditions, shear stress can promote nanoparticle aggregation by increas-

ing particle collisions during circulation, leading to a progressive increase in effective particle size under repeated flow exposure [5]. Furthermore, it is shown that nanoparticle aggregation under flow is a complex and accumulative process determined by parameters like shear conditions, particle properties, and circulation time, which making effective control harder in dynamic blood flow environments [6].

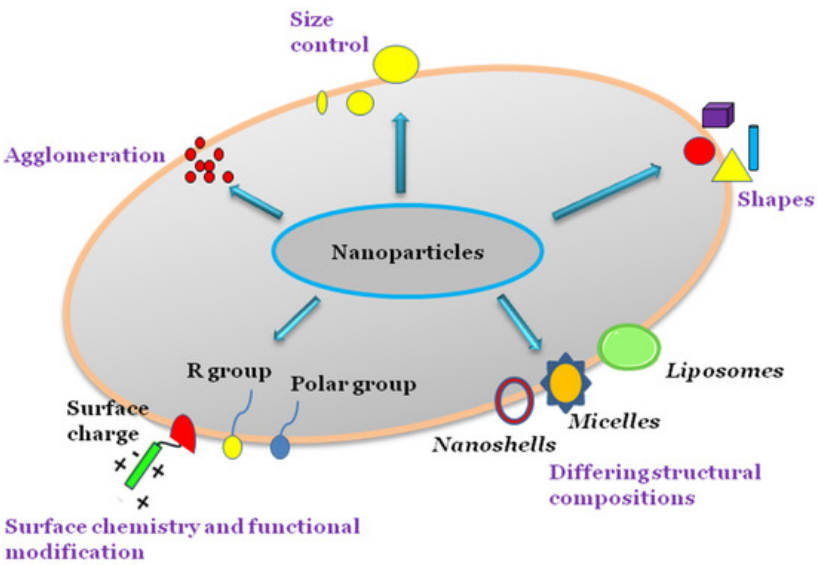

Figure 1: Physicochemical properties of nanoparticles [1]

What's more, shear force could also affect the structural stability of nanocarriers. Studies on lipid-based nanocarrier demonstrate that shear flow can induce gradual structural perturbations, such as deformation and interfacial rearrangement, without necessarily causing immediate particle rupture [7]. These effects become more pronounced under repeated flow exposure, leading to a progressive loss of encapsulation stability during circulation. As a result of structure instability, nanoparticles may exhibit early drug leakage within the bloodstream prior to reaching the target tissue [8]. This directly reduces the effective drug dose delivered to the tumour site. In summary, aggregation and sturctural instatbility significantly affect circulation-stage in nanocarrier-based drug delivery. These behaviours are inherently dynamic and accumulative, and therefore cannot be adequately captured by conventional static in vitro models. Hence, it is crutial for the development of flow-based experimental platforms that better reflect physiological circulation condition.

2.3 Existing Circulation-Mimicking Systems

Several in-vitro experimental platforms have been developed to reproduce characteristicsa of blood circulation for different applications. Researchers from Inha University developed a heart-pump-mimicking system as new vessel adhesives to stop bleeding from damaged vessels. The system simulates the rate of heart beat and strength of blood flow for standardized evaluation by adjusting pumping frequency and intensity, and is able to map several symptoms of circulation relevant to adhesive

testing [9]. In another system, an active mock circulation loop has been employed to investigate right-left flow balance concepts in total artificial heart designs, enabling dynamic data acquisition across varying hemodynamic conditions [10]. Efforts have also been made at the microscale, where a microfluidic blood capillary model in Fig. 2.was used to tailor the deformability behavior of soft hydrogel particles to mimic red blood cells, providing a platform for investigating bio-nano interactions under flow [11]. Compared with larger circulation systems, the microfluidic models offer more precise control over local flow conditions and particle deformation behaviors. However, each of these platforms focuses on specific aspects rather than addressing nanoparticle stability for target drug delivery.

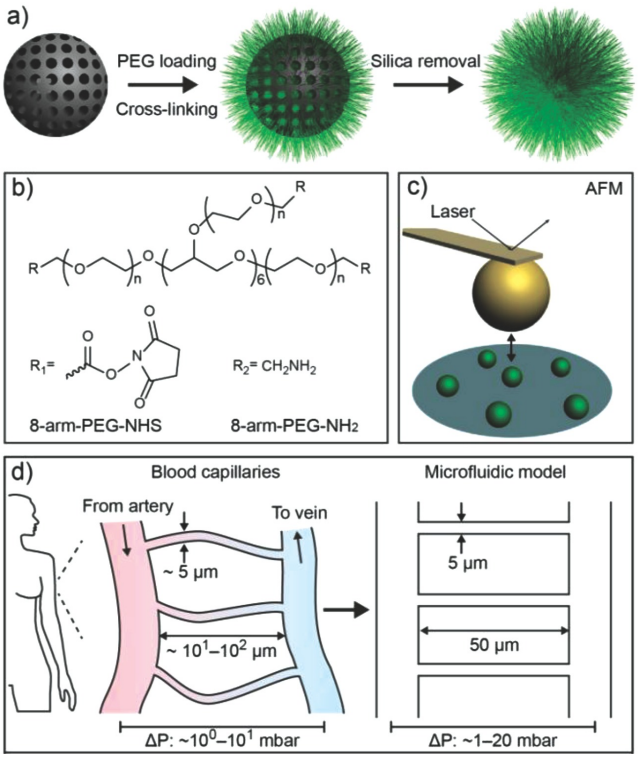

Figure 2: Microfluidic blood capillary model [11]

2.4 Microfluidic Circulation Approach

Microfluidics is a technology that controls the transport of fluids at the micrometre scale through precisely engineered channels [12]. Microfluidics has been recently used in many aspects of drug development as it offers a novel strategy for preparing nanoparticles with controllable properties due to its advantages, such as the ability of handle samples at pixel level, rapid mass transfer, and short production time [13]. More importantly, nanoparticles can be reproducibly synthesized in microfluidic devices, and their size, shape, surface properties, structure, and other characteristics can be well controlled and adjusted [14].

In recent studies, microfluidic platforms have been applied to investigate drug transport and tissue interactions under dynamic flow. For example, microfluidics

combined with endothelial cells has been used to model the blood brain barrier, enabling quantitative investigation of drug diffusivity across vascular interfaces under controlled flow conditions [15]. Besides, microfluidic systems incorporating cardiac cells are developed for cardiotoxicity and cardio-therapeutic drug screening, demonstrating the ability of microfluidics to support long-term perfusion [16]. As discussed last section, microfluidic capillary-like structures have also been employed to study the deformation and flow behaviour of PEG hydrogel particles designed to mimic red blood cells [11].

These studies demonstrate the capability of microfluidic platform to simulate key characteristics of the blood circulation, including shear force, confined flow, and continuous perfusion. Though most existing studies focus on cellular response and particle mechanics, microfluidic systems provide a promising basis for developing experimental platforms that address the limitations of static in vitro models discussed by integrating controlled flow conditions with recirculation and particle analysis. Hence advancing the validation of nanocarrier-based targeted drug delivery.

3 Reference

[1] Yusuf, A., Almotairy, A. R. Z., Henidi, H., Alshehri, O. Y. Aldughaim, M. S. Nanoparticles as Drug Delivery Systems: A Review of the Implication of Nanoparticles' Physicochemical Properties on Responses in Biological Systems. *Polymers (Basel)* **15**, 1596 (2023).

[3] Bioavailability and Delivery of Nutraceuticals Using Nanotechnology - Huang - 2010 - Journal of Food Science - Wiley Online Library.

[4] Zhang, H. *et al.* Shear stress regulation of nanoparticle uptake in vascular endothelial cells. *Regen Biomater* **10**, rbad047 (2023).

[5] Lv, B. *et al.* Influence of shear forces on the aggregation and sedimentation behavior of cerium dioxide (CeO2) nanoparticles under different hydrochemical conditions. *J Nanopart Res* **18**, 193 (2016).

[6] 1.Shrestha, S., Wang, B. Dutta, P. Nanoparticle processing: Understanding and controlling aggregation. *Advances in Colloid and Interface Science* **279**, 102162 (2020).

[7] Karaz, S. Senses, E. Liposomes Under Shear: Structure, Dynamics, and Drug Delivery Applications. *Advanced NanoBiomed Research* **3**, 2200101 (2023).

[8] Wang, Y., Pisapati, A. V., Zhang, X. F. Cheng, X. Recent Developments in Nanomaterial-based Shear-sensitive Drug Delivery Systems. *Adv Healthc Mater* **10**, e2002196 (2021).

[9] Jung, S.-M., Chung, G. Y. Shin, H. S. Artificial controlled model of blood circulation system for adhesive evaluation. *Sci Rep* **7**, 16720 (2017).

[10] Diedrich, M. *et al.* Experimental investigation of right-left flow balance concepts for a total artificial heart. *Artificial Organs* **45**, 364–372 (2021).

[11] Cui, J. *et al.* Super-Soft Hydrogel Particles with Tunable Elasticity in a Microfluidic Blood Capillary Model. *Advanced Materials* **26**, 7295–7299 (2014).

[12] Andersson, H. Berg, A. V. D. Microfabrication and microfluidics for tissue engineering: state of the art and future opportunities. *Lab Chip* **4**, 98 (2004).

[13] Zhao, C.-X., He, L., Qiao, S. Z. Middelberg, A. P. J. Nanoparticle synthesis in microreactors. *Chemical Engineering Science* **66**, 1463–1479 (2011).

[14] Marre, S. Jensen, K. F. Synthesis of micro and nanostructures in microfluidic systems. *Chem. Soc. Rev.* **39**, 1183–1202 (2010).

[15] Wang, Y. I., Abaci, H. E. Shuler, M. L. Microfluidic Blood-Brain Barrier Model Provides In Vivo-Like Barrier Properties for Drug Permeability Screening. *Biotechnol Bioeng* **114**, 184–194 (2017).

[16] Zhang, Y. S. *et al.* Multisensor-integrated organs-on-chips platform for automated and continual in situ monitoring of organoid behaviors. *Proceedings of

the National Academy of Sciences* **114**, E2293–E2302 (2017).